Property Rental and Lease Management System

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Course : Database Management System



Problem Identification

Property rental management involves handling large volumes of information — owners, properties, units, tenants, leases, rent payments, deposits, inspections, maintenance, renewals and terminations. Managing these activities manually or through disconnected records creates serious operational challenges.

Data Integrity Risks

Duplication, missed rent dues, and incorrect lease tracking undermine reliable record-keeping.

Operational Inefficiency

Identifying vacant units and generating accurate reports requires excessive manual effort.

The Need

A centralized DBMS to organize rental information, maintain integrity, and support effective property management.

Existing System & Problems

Current Approach

1)Manual records, spreadsheets, separate files and informal communication
2)Property, tenant, lease and payment information maintained in isolation

Key Problems

1)Data duplication: Same information recorded in multiple places
2)**Manual errors:** Incorrect entries in rent, payments or lease details
3)**Lease tracking:** Active leases, expiry dates and renewals harder to monitor
4)**Vacancy management:** Identifying available units is time-consuming
5)**Payment tracking:** Pending rent and history require manual checking
6)**Limited reporting:** Income, tenant history and maintenance reports take extra effort

Proposed System

A Python-based web application connected to a MySQL database for centralized property rental and lease management.



Property & Unit Management

Track properties, units and availability in one place.



Lease & Tenant Management

Owner/tenant registration, lease creation and lifecycle tracking.



Rent & Deposit Tracking

Schedules, payments, deposits and outstanding dues.



Inspection & Reports

Maintenance requests, inspections, renewals, terminations and search reports.

PYTHON | FLASK | MYSQL | MYSQL CONNECTOR/PYTHON

Project Scope & Objectives

Coverage Areas

1)**Property Management:** Manage properties, units, availability and details

2)**Owner & Tenant Management:** Maintain owner info and register/manage tenant records

3)**Lease Management:** Create leases and track periods, renewals and terminations

4)**Rent & Payment Management:** Maintain schedules, deposits, payments and outstanding dues

5)**Inspection & Maintenance:** Record inspections and manage maintenance requests

6)**Reports & Search:** Vacancy, due-rent, lease-expiry, owner-income, tenant-history and maintenance reports

Objectives

1)Design a centralized relational database for efficient rental and lease management

2)Maintain organized records of owners, properties, units, tenants and leases

3)Track rent schedules, deposits, payments and outstanding dues

4)Monitor unit availability and lease status for effective allocation

5)Manage inspections, maintenance requests and lease lifecycle events

6)Generate meaningful reports for vacancies, rent dues, lease expiry and income

7)Provide a simple Python-based web interface for database access



Users & User Roles

Primary User

Property Manager / Agency Staff

— Main user managing day-to-day rental activities.

Responsibilities

1)Manage owner, property and unit information
2)Register and manage tenants
3)Create and manage lease agreements
4)Track rent, deposits and payments
5)Record inspections and maintenance requests
6)Process lease renewals and terminations
7)Search records and generate management reports

Future Users

Property Owner

View property, lease and income information

Tenant

View lease, rent and maintenance information

Functional Requirements

Property & Unit Management

Add, update, delete and search properties and rental units

Owner & Tenant Management

Maintain owner details and register, update and search tenant information

Lease Management

Create leases and manage active, renewed and terminated leases

Rent & Payment Management

Maintain rent schedules, record payments and track outstanding dues

Inspection & Maintenance

Record inspections and manage maintenance requests

Search & Reports

Generate vacancy, rent, lease, income, tenant and maintenance reports

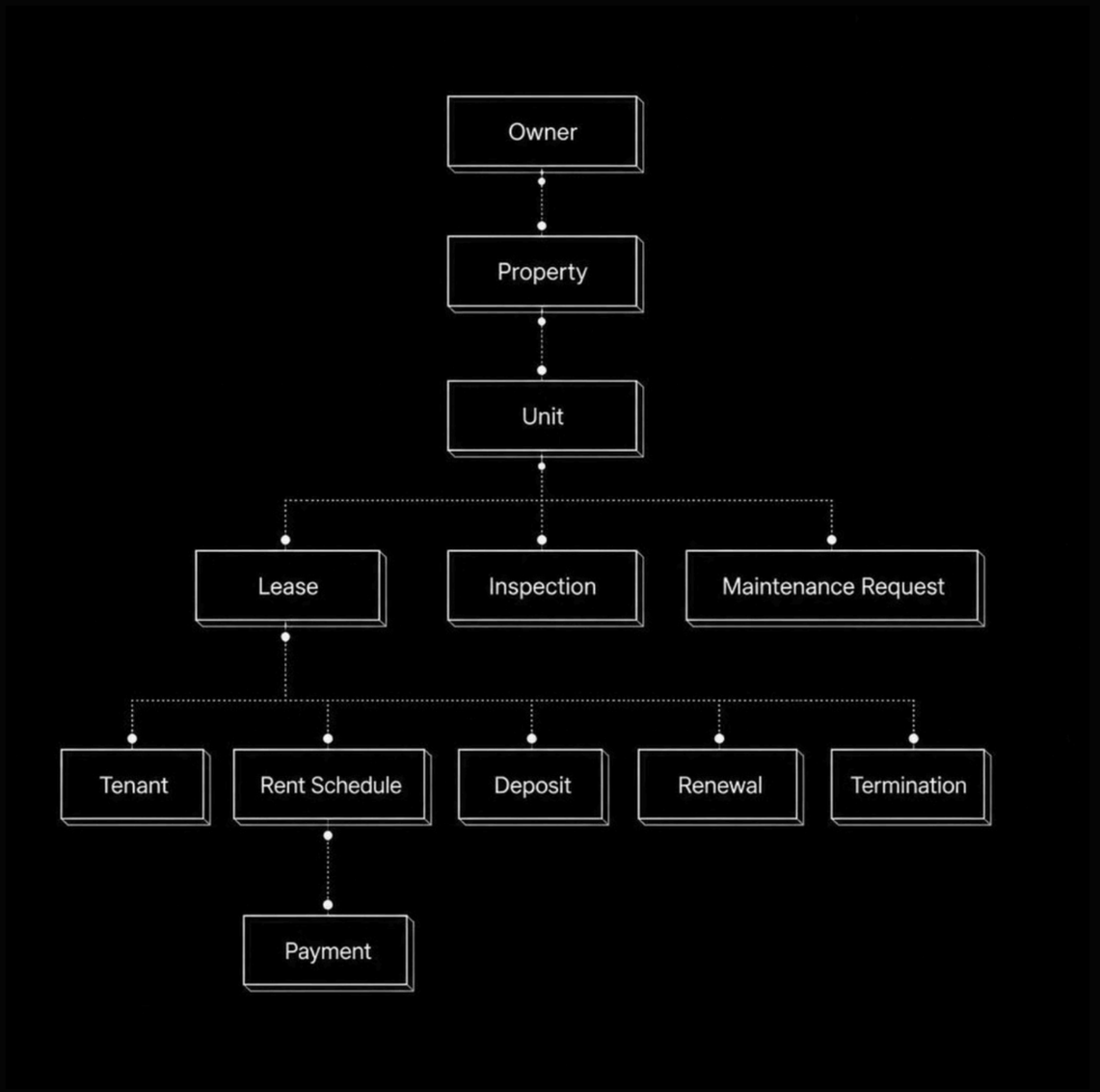
Major Entities

Core Data Entities

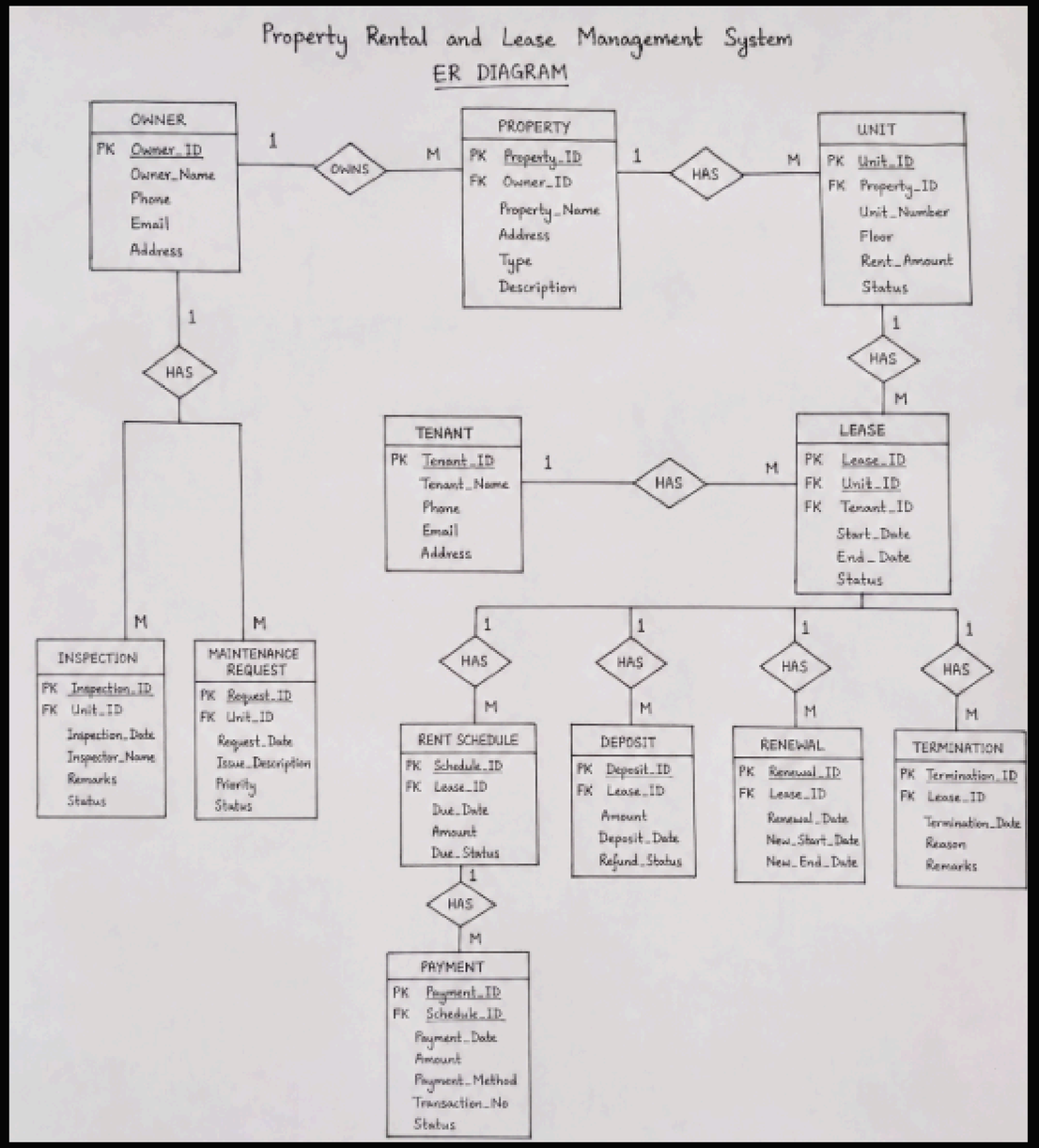
1)**Owner:** Property owner details
2)**Property:** Property information
3)**Unit:** Individual rental unit details
4)**Tenant:** Tenant information
5)**Lease:** Lease agreement details
6)**Rent Schedule:** Scheduled rent and due dates

Supporting Entities

1)**Deposit:** Security deposit details
2)**Payment:** Rent/payment transactions
3)**Inspection:** Inspection records
4)**Maintenance Request:** Issues and status
5)**Renewal:** Lease renewal details
6)**Termination:** Lease termination details



ER Diagram & Relationships



INITIAL RELATIONAL SCHEMA

Proposed Database Tables

Table Name	Primary Key (PK)	Important Foreign Key(s) (FK)
Owner	Owner_ID	—
Property	Property_ID	Owner_ID → Owner(Owner_ID)
Unit	Unit_ID	Property_ID → Property(Property_ID)
Tenant	Tenant_ID	—
Lease	Lease_ID	Unit_ID → Unit(Unit_ID), Tenant_ID → Tenant(Tenant_ID)
Rent_Schedule	Schedule_ID	Lease_ID → Lease(Lease_ID)
Deposit	Deposit_ID	Lease_ID → Lease(Lease_ID)
Payment	Payment_ID	Schedule_ID → Rent_Schedule(Schedule_ID)
Inspection	Inspection_ID	Unit_ID → Unit(Unit_ID)
Maintenance_Request	Request_ID	Unit_ID → Unit(Unit_ID)
Renewal	Renewal_ID	Lease_ID → Lease(Lease_ID)
Termination	Termination_ID	Lease_ID → Lease(Lease_ID)

CONCLUSION

The proposed Property Rental and Lease Management System provides a centralized approach to managing property rental activities. The project requirements, system scope, objectives, users, functional requirements and major database entities have been identified. The ER model and initial relational schema establish the foundation for the database implementation.

FUTURE SCOPE

1)Online Rent Payments – Enable tenants to make rent payments directly through the system.

2)Automated Notifications – Send reminders for upcoming rent dues, lease expiry and maintenance updates.

3)Owner & Tenant Portals – Provide separate access for owners and tenants to view relevant information.

4)Advanced Reports & Analytics – Add dashboards for rental income, occupancy, payments and maintenance trends.

5)Mobile Application – Extend the system to Android/iOS for convenient access.

6)Cloud Deployment – Deploy the application and database online for secure access from different locations.

7)Enhanced Security – Implement stronger authentication, authorization and data protection mechanisms.